



OPEN Promoting sustainable urban tourism in Historic Sari, Iran through GIS-based walkability planning

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Despite tourist attractions such as valuable historical buildings and axes, the historical district of Sari City has failed to attract visitors. By utilizing GIS route optimization and modeling exclusive pedestrian networks, this article seeks to improve walkability and promote tourism. This study concentrates on the Iranian city of Sari and emphasises sustainable urban development and tourism in urban planning. By incorporating the inverse desirability index of each network edge as a cost factor, an optimised tourist route is created that connects and preserves cultural heritage while efficiently connecting historical locations. The research demonstrates the applicability of this methodology to historical cities with comparable characteristics, providing an innovative approach to sidewalk modeling and accessibility analysis. Utilizing a multi-criteria GIS-based approach substantially enhances walkability in historic urban environments, according to the article's findings. The conclusion of the study highlights the creation of a more efficient tourist route in Sari, Iran, as well as its potential for boosting tourism and promoting sustainable urban communities in other historical areas.

Keywords Walkability, Pedestrian, Tourism, GIS, Route optimization, Historical cities, Cultural heritage preservation

Historic preservation strategies need to be tailored to the unique characteristics of heritage sites¹. The preservation of cultural and historical heritage, along with the development of walkable smart communities, enhances neighborhood livability^{2–4}. Among the priorities of Sari, Iran, which was designated as the ECO tourism capital in 2022, there is a priority on sustainable urban development and tourism promotion. The historical center of Sari is renowned for its rich urban texture, reflecting a blend of traditional Iranian architecture and vibrant marketplaces. It has witnessed the rise and fall of civilizations, leaving behind a tapestry of architectural gems and cultural heritage⁵. Exploring this captivating historical center offers an opportunity to delve into its unique characteristics and develop strategies for prioritizing tourist routes that showcase the city's rich cultural legacy.

Walkability Plays a significant role in a city's health, sustainability, and economic vitality^{6–8}. A walkable city ensures a safe and engaging pedestrian environment^{9,10}. Sari, with its concentration of tourist attractions in the old central zone, faces challenges in improving pedestrian infrastructure within its historical context¹¹.

Urban planning efforts in Sari prioritize pedestrian access and walkability by reducing car usage, creating pedestrian routes, and minimizing crossing wait times. Evaluating walkability in Sari is crucial to ensure effective strategies for pedestrian safety and accessibility.

Walkability in historical cities is a global concern. Many historical cities have experienced a decline in livability due to the prioritization of vehicles over pedestrians, resulting in negative impacts and the abandonment of city centers¹². However, there is a growing recognition of the importance of walkability in promoting sustainable urban development and enhancing the quality and attractiveness of urban areas¹³. Studies have shown that improving walkability can have multiple benefits, including improved health conditions, increased opportunities for citizens, and enhanced safety perception and attractiveness of historic city centers¹⁴. Various approaches

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and strategies have been proposed to enhance walkability, including the assessment of physical attributes, the identification of areas for improvement, and the implementation of pedestrian-friendly measures^{15,16}. The aim is to create pedestrian-friendly environments that promote walking activities and facilitate connectivity among houses, services, public facilities, and green spaces. However, it is important to consider the specific characteristics and challenges of each historical city when developing walkability indicators and interventions¹⁷. Walkability in historical cities has been a neglected topic in the study of transport and mobility¹². Many ancient cities around the world have experienced a decline in livability due to the prioritization of vehicles over pedestrians, leading to the abandonment of city centers^{16,18}.

Policies that promote pedestrianization contribute to sustainable urban development and climate change adaptation, while also encouraging walking and reducing car usage¹⁹. However, inefficient route planning and inadequate pedestrian infrastructure hinder tourist experiences in Sari, affecting walkability and urban growth. To address these challenges, a scientific approach utilizing GIS-based multi-criteria analysis and route optimization techniques can enhance walkability, optimize tourist routes, and improve urban development in Sari's historical core.

The historical center of Sari, despite its historical significance and potential as a tourist destination, faces challenges in attracting visitors due to inadequate pedestrian infrastructure²⁰ and tourist routes²¹. These shortcomings hinder the effective highlighting of the city's cultural heritage and fail to provide an optimal experience for tourists. To address this issue, there is a need for a comprehensive investigation into improving walkability and the design of tourist routes in Sari's historical center. This case study of Sari examines its historical context, focusing on 118 census blocks and various data variables. The study proposes a novel strategy that employs GIS route optimization to enhance walkability, preserve cultural heritage, and promote tourism. By creating an optimized tourist route that connects historic sites, encourages pedestrian activity, and preserves cultural heritage, this research analyzes sidewalk and crosswalk access through GIS modeling. This paper focuses specifically on developing an optimized pedestrian routing methodology rather than surveying areas currently lacking walkability. The aim is to improve walkability and attract tourists in historical contexts, with wider potential for implementation in similar cities, fostering sustainable and inclusive urban communities. The study's significance is accentuated by addressing key research questions:

RQ1. How can GIS-based walkability modeling improve tourism accessibility in historical urban areas?

RQ2. What walkability indicators should be integrated to develop efficient, pedestrian-friendly tourism routes?

RQ3. What are the potential benefits of integrating GIS route optimization and exclusive sidewalk network modeling for improving the tourist experience and preserving cultural heritage?

Literature review

Pedestrian infrastructure is crucial for accessing public spaces and serves as the primary mode of transportation^{22–24}. These networks consist of interconnected paths exclusively for pedestrians or pedestrians on small wheels²⁵. However, historic cities often have narrow, winding streets that pose challenges to walkability²⁶.

Using GIS route optimization can be beneficial for improving the walkability index in historical cities^{15,27}. This approach helps planners and designers identify efficient walking routes^{28–30}. It has been demonstrated in previous studies that pedestrian infrastructure enhances the perception of safety and comfort, encouraging more walking³¹. However, several factors affect the walking experience, including the built environment and streetscape^{32–34}.

There are multiple factors that affect pedestrians' perceptions of comfort, safety, and connectivity. In determining their level of inclination toward walking and perception of walking as an efficient mode for transportation, these factors are vital^{35–37}.

Over the past few decades, sustainable tourism research has grown significantly³⁸. Policies to promote pedestrianization can contribute to sustainable urban development and climate change adaptation, while encouraging walking and reducing the use of cars¹⁹. These policies can help preserve existing values and create more walkable and livable cities.

Theories explaining walking behavior

Volition and intention, specifically shaped by perception and subjective norms, play an important role in determining walking behavior, according to the theory of reasoned action (TRA)³⁹. In addition to TRA, TPB (theory of planned behavior) considers attitude, normative expectations, and perceived behavioral control as determinants of behavior⁴⁰. TPB suggests that enhancing walkability can be achieved by promoting positive attitudes, supportive social norms, and improved pedestrian infrastructure. The Expectancy-Value Theory (EVT) highlights two crucial factors—expectancy (belief in the ability to perform a behavior) and value (the significance attached to behavior outcomes)—in motivating walking behavior⁴¹. By fostering expectancy through skill development and resources, as well as emphasizing the benefits of walking, engagement in walking activities can be increased. While research on EVT in walkability is limited, understanding these theories provides valuable insights for influencing walking behavior and informing strategies to promote walking behaviors^{42–46}.

Walkability in historical cities and tourism

Walkability is a key component of sustainable urban tourism across multiple contexts. Scott and Cooper highlight walking trails as a 'front room' sustainability innovation⁴⁷. Priatmoko et al. show how placemaking can create walkable spaces and pedestrian infrastructure to enable slow tourism in unpopular urban destinations⁴⁸. Similarly, Grube discusses how Berlin promotes walkability through local transport and pedestrian-friendly infrastructure as part of its "city-compatible tourism" approach. In various cases, developing walkable urban

spaces through placemaking helps support sustainable transportation choices and tourism activities⁴⁹. Studies focusing on increasing tourism in historical cities have explored the factors that contribute to walkability and those that hinder it. A comparative study of Kyoto, Japan, and Xi'an, China, revealed that grid-based street networks and sustainable development can enhance walkability in fragmented historical blocks¹². Another study examined the impact of Singapore's city planning and transportation policies on the walking experience of tourists²⁰. Additionally, a comprehensive review of walkability studies identified over 2400 journal articles⁵⁰. Furthermore, specific historical cities like Rome-Vatican and Venice have been examined to identify their walkability weaknesses and propose solutions^{20,51}.

Previous research utilizing the Theory of Planned Behavior (TPB) in Iran has yielded valuable insights, shedding light on behavioral dynamics, yet these studies are not without limitations. The utilization of self-reported data introduces the potential influence of social desirability bias, rendering it inadequate in accurately reflecting individuals' actual behavior. Moreover, the concentration on specific target groups confines the generalizability of findings. To bolster the efficacy of TPB, the incorporation of supplementary predictors, such as moral norms and self-identity, is proposed, particularly in elucidating behaviors pertaining to organic food consumption. The consideration of diverse demographic characteristics and cultural factors assumes significance in comprehending behavioral intentions within the Iranian context. Furthermore, prevailing research predominantly emphasizes pro-environmental behaviors, inadvertently overlooking other pertinent domains. Future investigations should explore the applicability of TPB to health-related behaviors and consumer choices. Despite these limitations, TPB research continues to inform our understanding of behavioral intentions, and addressing these gaps and constructing culturally relevant models will facilitate more effective interventions and policy-making endeavors in Iran.

Proposed parameters and sub-parameters for walkability evaluation

Enhancing walkability in historical cities contributes to a more enjoyable experience for visitors. However, achieving walkability involves considering the unique characteristics and challenges of each city, such as limited services and high tourist volumes. Sustainable development and urban planning play a crucial role, but their implementation varies across cities. To optimize walkability for tourists, this research conducted a comprehensive evaluation of the walkability index by reviewing multiple articles. Parameters and sub-parameters from 15 studies were synthesized, prioritizing those with frequent occurrence. A hierarchical table of parameters was established

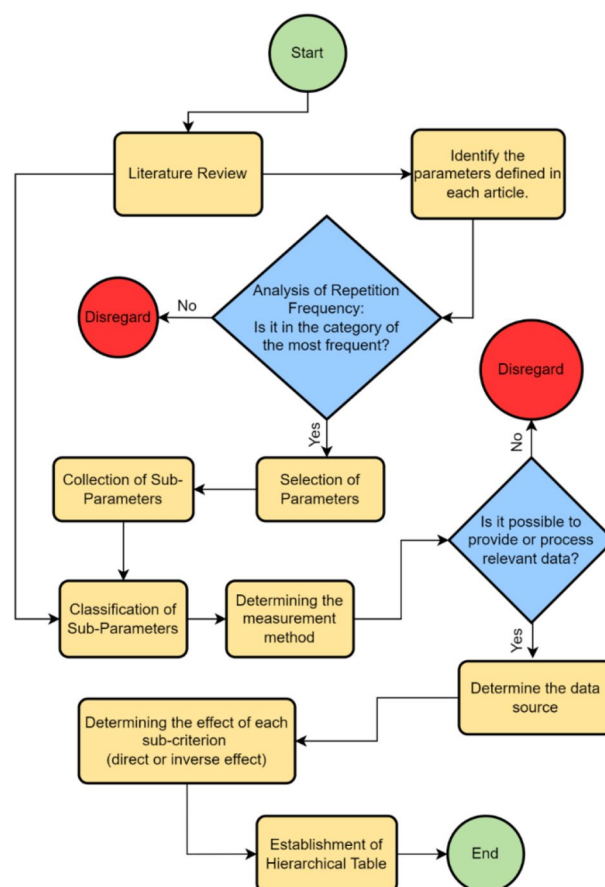


Figure 1. Methodological approach for parameter and sub-parameter selection.

to systematically analyze the significant factors influencing pedestrian activity (Fig. 1). This methodology ensures a systematic assessment of parameters and sub-parameters in the pedestrian environment.

Table 1 presents a comprehensive set of parameters categorized into five main categories (attractiveness, accessibility, safety, comfort, and legibility) for evaluating walkability in urban environments. Each category includes several sub-parameters with their respective evaluation measures.

The process depicted in the flowchart involves several important steps. It begins with the Literature Review phase, where the collection and review of relevant articles take place. The next step is the Identification of Sub-Parameters, where sub-parameters are identified based on the information gathered from the reviewed articles. Following that is the Selection of Parameters step, where the main parameters of interest are determined. In the Categorization of Sub-Parameters phase, the identified sub-parameters are grouped or classified based on their characteristics. The next step, Determining the Measurement Method, involves establishing the appropriate method for measuring the parameters. Moving forward, the Analysis of Repetition Frequency step assesses whether the parameters fall under the most frequent category. If they do, the process proceeds to the Disregard step, where those parameters are disregarded. Finally, in the Establishment of Hierarchical Table phase, a hierarchical table is created for the remaining parameters. These consecutive phases establish a conceptual structure for the procedure, facilitating enhanced comprehension of the entire progression. The structured framework enables a systematic and holistic assessment, considering multiple aspects that contribute to pedestrian-friendly environments. Additionally, Table 2 provides suggested evaluation measures specifically designed for the sub-parameters in walkability assessment, further supporting the comprehensive evaluation process.

Data and methods

Data

Sari, the capital of Mazandaran province in northern Iran, is a culturally significant city known for its rich history¹⁰³. The central part of the city, covering approximately 150 hectares, serves as the main hub of activities and holds great tourism potential. However, the city faces challenges with its pedestrian infrastructure, hindering the full utilization of its historical and cultural assets for tourism¹⁰⁴. Enhancing the pedestrian infrastructure could further unlock the city’s tourism potential.

The GIS data utilized in this study was sourced from the Municipality of Sari and the National Cartographic Center of Iran (base map scaled 1:2000). It comprises vector layers of building footprints, road networks, land use, and points of interest, along with raster data such as digital elevation models and satellite imagery. The spatial accuracy of the vector data is within 2 m, and the resolution of the raster data is 1 m.

Figure 2 depicts the geographic position of the study area situated in the central region of Sari City, Iran. It provides an overview of the specific area under investigation within the city’s boundaries. In addition, Fig. 2 illustrates the historical context area, highlighting the spatial distribution of tourism attractions within the studied region. This map offers valuable insights into the concentration and dispersion of significant historical landmarks and sites of tourist interest.

This paper focuses on a specific area in the center of Sari city to analyze the pedestrian network’s walkability. The selected area covers part of the central core, spanning 194 hectares and situated at an average distance of 700–1100 m from the central point. It features a compact urban structure with diverse spaces and streets, including residential, commercial, public service areas, green spaces, cultural and tourism destinations. Within this area, there is a network of 90 streets that are further divided into 118 paths, squares, and pedestrian routes. The total length of this network measures 27.8 km. Its proximity to essential amenities and pedestrian hubs in the surrounding areas, such as the central municipality, stadium, hospital, hypermarkets, bus and train stations, and historical buildings, makes it an intriguing site for study (Fig. 2).

The attractions in this area offer various points of interest along a walkability route. Visitor-Focused Urban Squares like Saat Tower and Qaaran Street provide vibrant urban experiences for visitors. Heritage Sites such as Fazeli House, Remedani House, Vaziri Bathhouse, Sardar Jalil Historical House, Sultan Zeynolabedin Tower, and Sisdahpich Alley showcase the city’s rich historical and architectural heritage. Religious Heritage Centers, including Ab-Anbarno Mosque, Mustafa Khan Mosque, Jameh Mosque of Sari, and Imam Zeynolabedin Mosque, hold spiritual and cultural significance. Cultural Heritage Centers like The historical house of Mirgati, The historical house of Kolbadi, and Sarigraphi Centre preserve and exhibit the region’s cultural heritage. Commercial Centers such as Ebne ShahrAshoub Bazaar, Bazaar Rooz, Mofidi Shopping Center, Elahieh Shopping Center, and Nargesieh Bazaar offer shopping and commercial opportunities. Urban Parks like Velayat Park and

Parameters	Description	References
Attractiveness	Examines neighborhood land use diversity, amount of green space, quality of public spaces, presence of retail, and streetscape design. Factors like mix of uses, parks, plazas, benches, and pedestrian amenities enhance attractiveness	51–54
Accessibility	Evaluates sidewalk continuity, width, obstacles, proximity to parking and transit. Wider sidewalks with few barriers that connect to destinations improve accessibility	51,54–56
Safety	Considers lighting, traffic volume/speed, presence of crosswalks and sidewalks. More lighting, lower speeds, frequent crossings, and continuous sidewalks enhance pedestrian safety	57–59
Comfort	Measures availability of shade, shelters, noise levels, odors. More trees, awnings, low noise, and pleasant smells improve comfort	57,60–63
Legibility	Evaluates street network connectivity and building enclosure. More small blocks and consistent building walls make places easier to navigate	34,64–67

Table 1. List of key parameters for evaluating walkability.

	Sub-parameters	Measure	Walkability classification	Source
Attractiveness	Land use-Mix: Blending of diverse land uses in an area, enhancing pedestrian appeal and promoting active travel ^{68,69}	Activity density (non-residential land use density)	Low → High	Audit/GIS
	Access to urban green spaces with benefits like promoting physical activity, improving air quality, and enhancing well-being ^{70,71}	Urban green space density (parks, trees, boulevards density)	Low → High	Audit/GIS
	Public space: Quality public spaces improve public health, support local economies, and enhance the urban environment ^{59,72,73}	Urban public space density	Low → High	Audit/GIS
	Density of retail shops along pedestrian routes attracts visitors and enhances the route's attractiveness ^{74,75}	Retail shop density adjacent to streets	Low → High	GIS
	Street Design Furniture: Condition of sidewalk pavement affects pedestrian comfort, safety, and accessibility ^{76,77}	Density of urban furniture installations	Low → High	Audit
Accessibility	Sidewalk Pavement: Condition of sidewalk pavement affects pedestrian comfort, safety, and accessibility ^{62,78}	Pavement quality	Low → High	Audit
	Sidewalk width: evaluating sidewalk width is important for accommodating diverse users and promoting interaction ⁷⁹	Adequate sidewalk width	Low → High	Audit/GIS
	Obstacles: Identifying and addressing obstacles like street furniture and parked cars is crucial for accessibility ^{80,81}	Physical barrier density	Low → High	Audit
	Proximity to public parking lots: Availability of public parking lots encourages a park-once-walk mentality and increases walkability ^{82,83}	Proximity to public parking lots	Long distances → Short distances	GIS
	Proximity to public transportation: Access to public transportation enhances walkability, reduces traffic congestion, and promotes active modes of transportation ^{84,85}	Proximity to taxi and bus stations	Long distances → Short distances	GIS
Safety	Proximity to police stations: Nearby police stations enhance pedestrians' perception of safety and encourage walking ^{86,87}	Proximity to police guard booths	Long distances → Short distances	Audit
	Lighting system: Street lighting improves pedestrian safety, visibility, and well-being ^{88,89}	Streetlight density	Low → High	Audit
	Proximity to crosswalk: Availability of crosswalks enhances pedestrian maneuverability and influences route choices ^{90,91}	Proximity to pedestrian crossings	Long distances → Short distances	GIS
	Traffic management: Proper management of pedestrian and bicycle traffic flow enhances safety ^{92,93}	Pedestrian and bicycle traffic flow mixing intensity	Low → High	GIS
	Intersection density: Pedestrian safety is influenced by the intensity of mixing between pedestrians and vehicles ⁹⁴	Weighted and hierarchical intersection density	High → Low	GIS
Comfort	Shade: Availability of shaded areas, particularly from trees, improves pedestrian comfort ^{63,95}	Sunlight peak shadow space	Low → High	Audit/GIS
	Shelter: Covered sidewalks provide shelter from rain and snow, enhancing comfort ⁹⁶	Sheltered space density	Low → High	Audit/GIS
	Noise: Density of busy and noisy areas negatively affects pedestrian comfort and walkability ⁹⁷	Congestion at high traffic intersections and noisy commercial areas	Short distances → Long distances	Audit/GIS
	Smell: Avoidance of unpleasant odor-emitting activities contributes to pedestrian comfort ⁹⁸	Avoidance of unpleasant odor-emitting activities (such as fish and butcher shops)	Short distances → Long distances	Audit
Legibility	Permeability: Ability of urban blocks to accommodate pedestrians and vehicles, influencing walkability and urban accessibility ^{99,100}	Block permeability level (inverse relationship with block size)	Low → High	GIS
	Enclosure: Influence of factors like block length, edge conditions, and geometric attributes on the perception of walkable neighborhoods ^{93,101,102}	Average adjacent building height to street width ratio	High → Low	Audit/GIS

Table 2. Suggested evaluation measures for sub-parameters in walkability assessment.

Shahrdari Park provide open green spaces for relaxation and recreational activities. These tourism attractions in Sari cater to diverse interests by combining historical, cultural, religious, commercial, and natural elements to provide a rich visitor experience.

Methods

To enhance walkability through GIS route optimization, the initial step involves gathering data on the city's infrastructure, such as sidewalks, crosswalks, pedestrian signals, and street lighting¹⁰⁵. This data is analyzed using digital mapping techniques to assess the current state of walkability. ArcGIS Pro 3.2 (<https://www.esri.com/en-us/arcgis/products/arcgis-pro/overview>) was used to create the maps presented in Figs. 2, 5, 6, 7, 8, and 9.

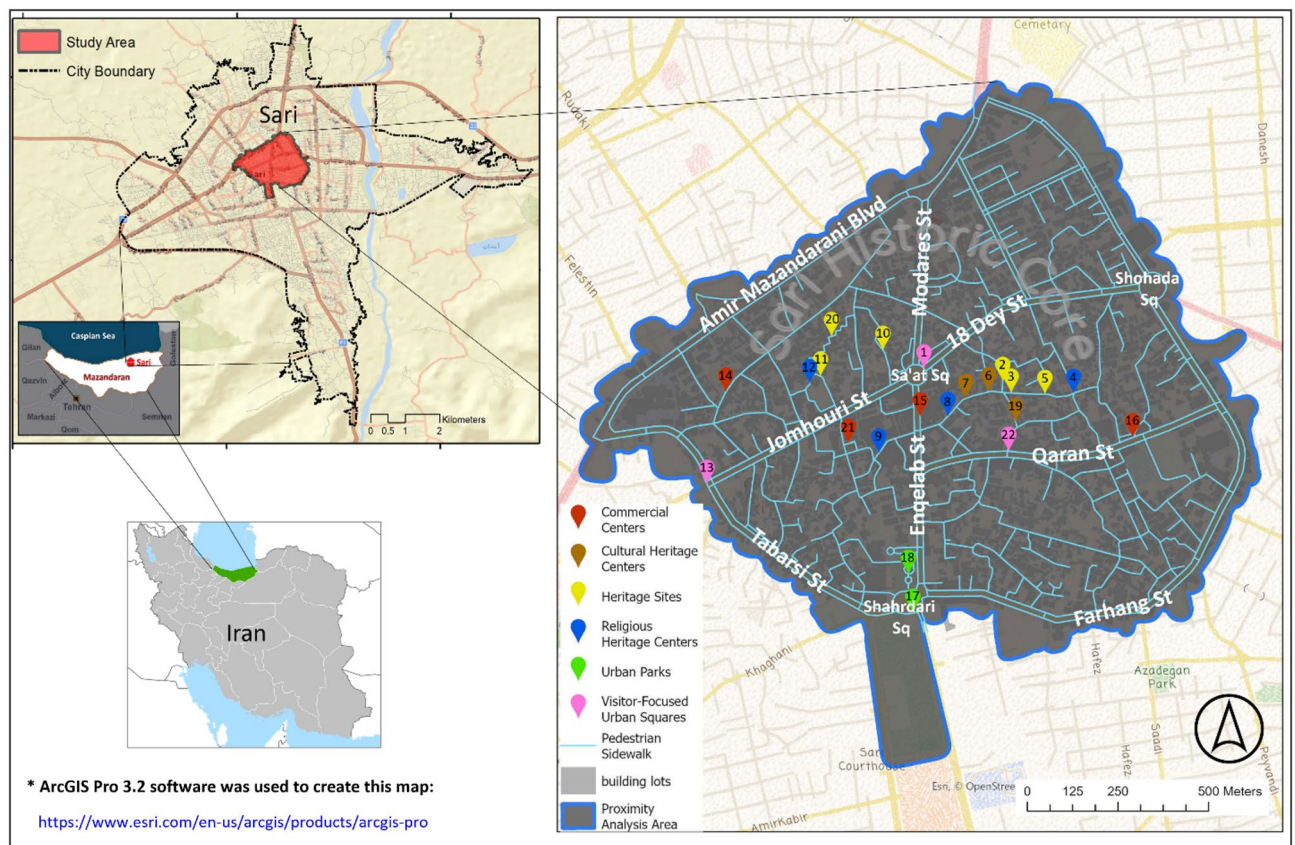


Figure 2. Historical context area of Sari, Iran (Hierarchical location maps depicting (a) Iran country, (b) Mazandaran province, (c) Sari city boundary, and (d) Study area encompassing the historical core of Sari city. Labeled symbols indicate the distribution of cultural, religious, commercial, and natural attractions analyzed for tourism walkability mapping, including: (1) Saat Tower, (2) Fazeli House, (3) Remedani House, (4) Ab-Anbarno Mosque, (5) Vaziri Bathhouse, (6) The historical house of Mirgati—Museum, (7) The historical house of Kolbadi- Museum, (8) Mustafa Khan Mosque, (9) Jameh Mosque of Sari, (10) Sardar Jalil Historical House, (11) Sultan Zeynolabedin Tower, (12) Imam Zeynolabedin Mosque, (13) Ebne ShahrAshoub Bazaar, (14) Bazaar Rooz, (15) Mofidi Shopping Center, (16) Elahieh Shopping Center, (17) Velayat Park, (18) Shahr-dari Park, (19) Sarigraphi Centre, (20) Sizdahpich Alley, (21) Nargesieh Bazaar, (22) Ghaaran Street.)

Using GIS route optimization, areas in need of improvement are identified by evaluating factors like missing or damaged sidewalks, unclear crosswalks, and poorly timed pedestrian signals¹⁰⁶. Street conditions, intersections, and pedestrian traffic patterns are considered to determine the most optimal walking routes.

Once the areas for enhancement are identified, GIS route optimization facilitates the development of a plan to improve the walkability index in historical cities. Strategies may involve adding sidewalks, improving crosswalks, and optimizing pedestrian signals¹⁰⁵, with the aim of creating a safe, efficient, and enjoyable walking environment.

To assess walkability and establish the walkability index, various data inputs are utilized, including zoning data, building footprints, street network polygons, census data, and an urban GIS-based audit GDB. Incorporating urban transportation planning documents and employing the ArcScan process allows for adjustments to the pedestrian networks. The methodology involves modeling pedestrian axes, conducting spatial analysis, reclassifying rasters, spatially joining data, standardizing values, applying AHP-based weights, and building a pedestrian network dataset for route optimization.

To accurately represent pedestrian perceptions and movement, the pedestrian network is modeled using street network and pedestrian data, such as sidewalks and squares, following Zhang's approach¹⁰⁷. The methodology includes modeling pedestrian axes, splitting lines at junctions, and utilizing spatial analysis techniques to evaluate 21 walkability parameters¹⁰⁸. The resulting classified rasters are reclassified on a scale of 1–10, representing walkability from lowest to highest. By spatially joining the reclassified raster values with pedestrian axes, values are standardized for consistency¹⁰⁹. The AHP (Analytical Hierarchy Process) method is used to compute the weights assigned to each walkability parameter, which are then multiplied by the standardized values to obtain the final walkability index (FWI). The reverse FWI values are invented as cost values, stored as the reverse final walkability index (RFWI). A pedestrian network dataset (PND) has been created, incorporating cost and length parameters for route optimization between tourism stops, facilitating the identification of optimal walking routes in historical contexts.

Multi-criteria evaluation

Multi-criteria evaluation (MCE) proves to be an invaluable tool in the realm of urban planning and design as it enables the consideration of multiple factors and prioritization of the most important ones^{110–112}.

In walkability studies, the use of multi-criteria evaluations (MCAs) is widespread^{54,56,113}. This approach assesses the walkability of urban centers by integrating various criteria into a GIS-based assessment.

GIS plays a crucial role in this process by integrating and analyzing relevant data, allowing planners and designers to identify areas for improvement that promote sustainable transportation and walking^{52,61,114}. GIS-based walkability assessments provide valuable insights to urban planners, aiding in the formulation of policies that foster livable and healthy communities.

However, the effectiveness of MCE in developing sustainable and walkable cities through GIS-based walkability assessments relies on careful selection of criteria, data sources, and weighting.

This study developed the MCA in two main phases: attribute selection and evaluation, as well as attribute weighting. The following subsections describe the steps taken in these phases.

In the research conducted in Sari, walkability was evaluated using 21 Characteristics of the physical environment and streetscape, which were selected based on a thorough review of literature. These attributes were further assessed as Sub-parameters of six primary criteria that include Attractiveness, Accessibility, Safety, Comfort, and Legibility. The article explains the attributes utilized, the rationale for their selection, the methods employed for evaluation, and sources of data. A condensed version of this data is presented in Table 2.

AHP, developed by Saaty¹¹⁵, is an important technique for land suitability analysis¹¹⁶. It falls under the multi-criteria decision analysis approach and aids planners and decision makers in analyzing data for future land-use changes¹¹⁷. AHP involves creating a hierarchical structure, pairwise comparisons using a questionnaire to determine the weights assigned to parameters based on their relative importance (local weights). These weights are then synthesized to establish global priorities (weights) for the final decision¹¹⁸. AHP has been integrated with GIS tools to identify criteria importance and calculate weights based on expert opinions¹¹⁹.

AHP breaks down intricate problems into structured layers, assessing factors within each layer utilizing a scale ranging from 1 to 9, forming a matrix of judgments, and determining the largest eigenvalue and its corresponding eigenvector. Weights are determined step-by-step, layer by layer, until the cumulative weights of all factors from the lowest criterion to the ultimate goal are established¹²⁰.

Pairwise comparisons are made between parameters in AHP, determining their relative importance through dominance coefficients (a_{ij}). The Saaty fundamental scale, ranging from 1 to 9, is used to assign values for pairwise comparisons. The AHP weighting process calculates the geometric mean of expert evaluations in the comparison matrix, normalizing the results to ensure a sum of 1 for each criterion's weight. The matrix's consistency is evaluated by examining the consistency index and ratio, with the objective of achieving values below 0.1^{121,122}.

The AHP methodology involves the following steps^{***}:

i) Constructing a pairwise comparison matrix A , where each element a_{ij} represents the relative importance of criterion i over criterion j , based on Saaty's fundamental scale (1–9).

ii) Calculating the priority vector w , which represents the relative weights of the criteria, by finding the principal eigenvector of matrix A .

iii) Evaluating the consistency of the judgments using the Consistency Index (CI) and Consistency Ratio (CR):

$$CI = (\lambda_{\max} - n) / (n - 1).$$

where $\lambda_{\max}$ is the principal eigenvalue of matrix A , n is the number of criteria, and

$$CR = CI / RI.$$

RI is the Random Consistency Index, which is a constant value depending on n .

A CR value less than 0.1 is considered acceptable, indicating consistent judgments.

The weight factors (1–9) used in the AHP methodology were assigned based on expert opinions and a literature review. A group of urban planners and tourism experts from the Municipality of Sari and local universities were consulted to provide their judgments on the relative importance of the walkability criteria through pairwise comparisons. These judgments were then combined using the geometric mean to derive the final weight factors presented in the study. The weights obtained are shown in Fig. 3. The criterion rated the highest is Attractiveness (0.365), following that, the Accessibility parameter (0.264) and Safety (0.175) are ranked accordingly. Comfort receives the lowest evaluation (0.096). Furthermore, Fig. 3 illustrates the AHP decision tree, showcasing the assigned weights for both the parameters and sub-parameters. Additionally, Table 3 presents the Consistency Index (CI) along with Consistency Ratio (CR) values, which validate consistency.

The concluding stage of the study involved the computation of a comprehensive walkability index for the (walkable) network. This index was determined by combining the evaluation of each sub-parameter with the normalized weights derived from the Analytic Hierarchy Process (AHP). The analysis was conducted within a GIS environment using Esri's ArcGIS software. Following defining the parameters and assigning scores to them, the values of the six parameters were combined and subsequently re-normalized. A walkability map as the outcome representing a range of 0 to 1, enabling the classification of edges (walkable paths) based on the evaluated parameters. Figure 4 presents a flowchart illustrating the optimization process of the urban walkability index as a tourism route.

In Fig. 4, the flowchart outlines the process of calculating the Walkability Index, which involves various steps for analyzing urban parcels, buildings footprints, streets network, urban blocks, census data, zoning regulations, and urban GIS-based audit data. The process begins with data collection and proceeds to adjust pedestrian networks and gather preliminary data.

The next steps involve modeling pedestrian axes and crossings, integrating preliminary data with the model, joining required information to attribute of each axis, and designing a comprehensive geodatabase. Spatial analysis is performed to calculate parameters 1–21 in raster format, including Euclidean distance, Manhattan distance, and point density.

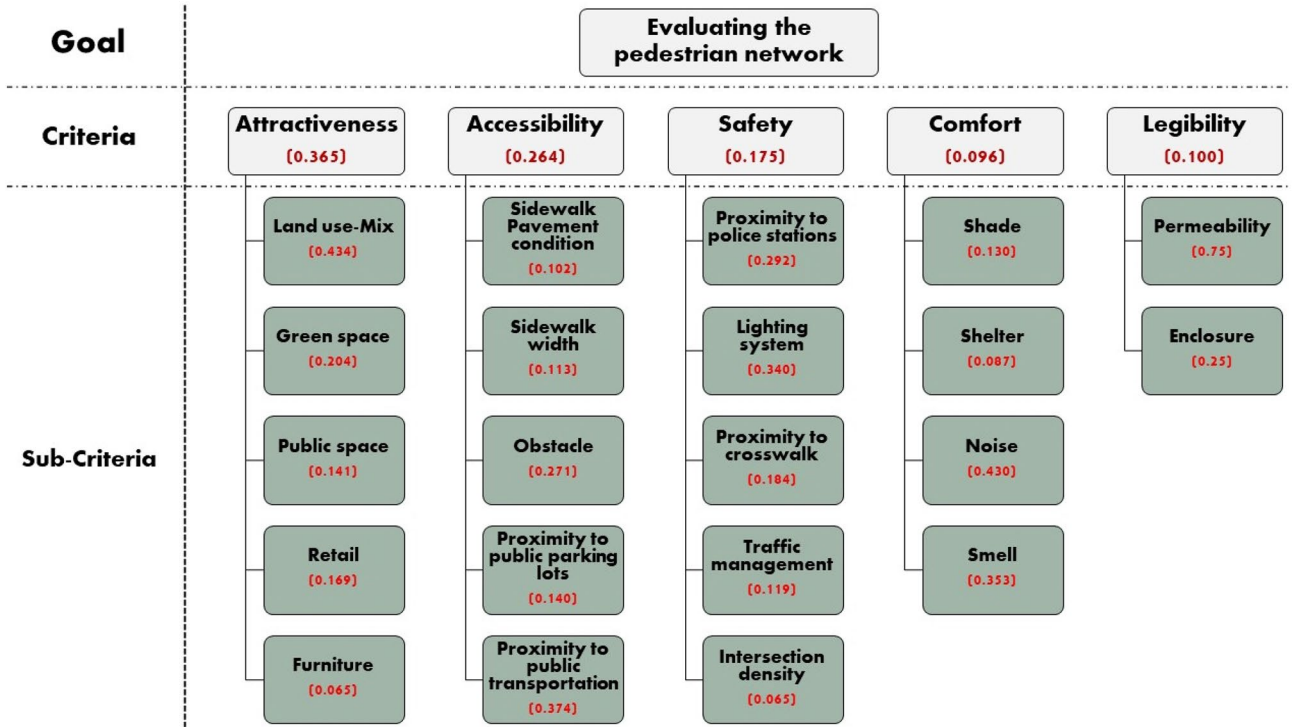


Figure 3. AHP decision hierarchy and weights for parameters and sub-parameters.

Sub-parameters and Parameters	λ_{max}	CI	Random CI	CR
Attractiveness	5.347	0.086	(n = 5), RI = 1.12	0.077
Accessibility	5.440	0.110	(n = 5), RI = 1.12	0.098
Safety	5.389	0.097	(n = 5), RI = 1.12	0.086
Comfort	4.065	0.021	(n = 4), RI = 0.90	0.024
All 5 parameters	5.194	0.048	(n = 5), RI = 1.12	0.043

Table 3. Validation of AHP using Consistency Index (CI) and Consistency Ratio (CR).

The flowchart then describes reclassifying the raster data based on a walkability scale and spatially joining the reclassified raster with pedestrian axes. The values of pedestrian axes are standardized, and AHP calculated weights are multiplied to the standardized weights. FWI is calculated by summing up the weighted standardized values, and the Reverse Final Walkability Index (RFWI) is obtained by reversing the FWI values.

The process concludes with the creation of a pedestrian network dataset (PND) based on the RFWI values and the definition of specific tourism stops within the pedestrian network. The Walkability Index and RFWI provide valuable insights for urban planning and decision-making processes.

Results

The evaluation of the five criteria within the study area is illustrated in Fig. 5, providing a visual representation of their respective assessments. Meanwhile, the evaluation of the 21 sub-parameters can be found in Fig. 11 in the Appendix, where a series of maps display their performance. In these maps, streets receiving the highest scores are depicted in green, while those with the lowest scores are indicated in red. Specifically, Fig. 6 showcases the overlapping sub-criteria and value recategorization maps, offering insights into the integration and synthesis of these criteria for each individual criterion.

To comprehensively evaluate pedestrian conditions in the study area, the normalized values resulting from the assessment were integrated with the weights derived using the AHP. This integration process, performed in GIS, yielded the outcomes displayed in Fig. 6. This map illustrates the final weighted overlap, showcasing the combined influence of the five key criteria in the assessment of walkability. Additionally, Fig. 7 presents the optimal walking route based on the proposed methodology, highlighting the most favorable path for pedestrians within the study area. The route optimization algorithm uses the impedance values to select the optimal path that minimizes the total impedance. Edges with lower RFWI scores were prioritized, resulting in a route that maximizes walkability by avoiding difficult edges with high impedance.

The optimized pedestrian network dataset was integral to determining the ideal walking route by enabling impedance-based optimization functions, as outlined in “Methods”. Optimization tools could select optimal

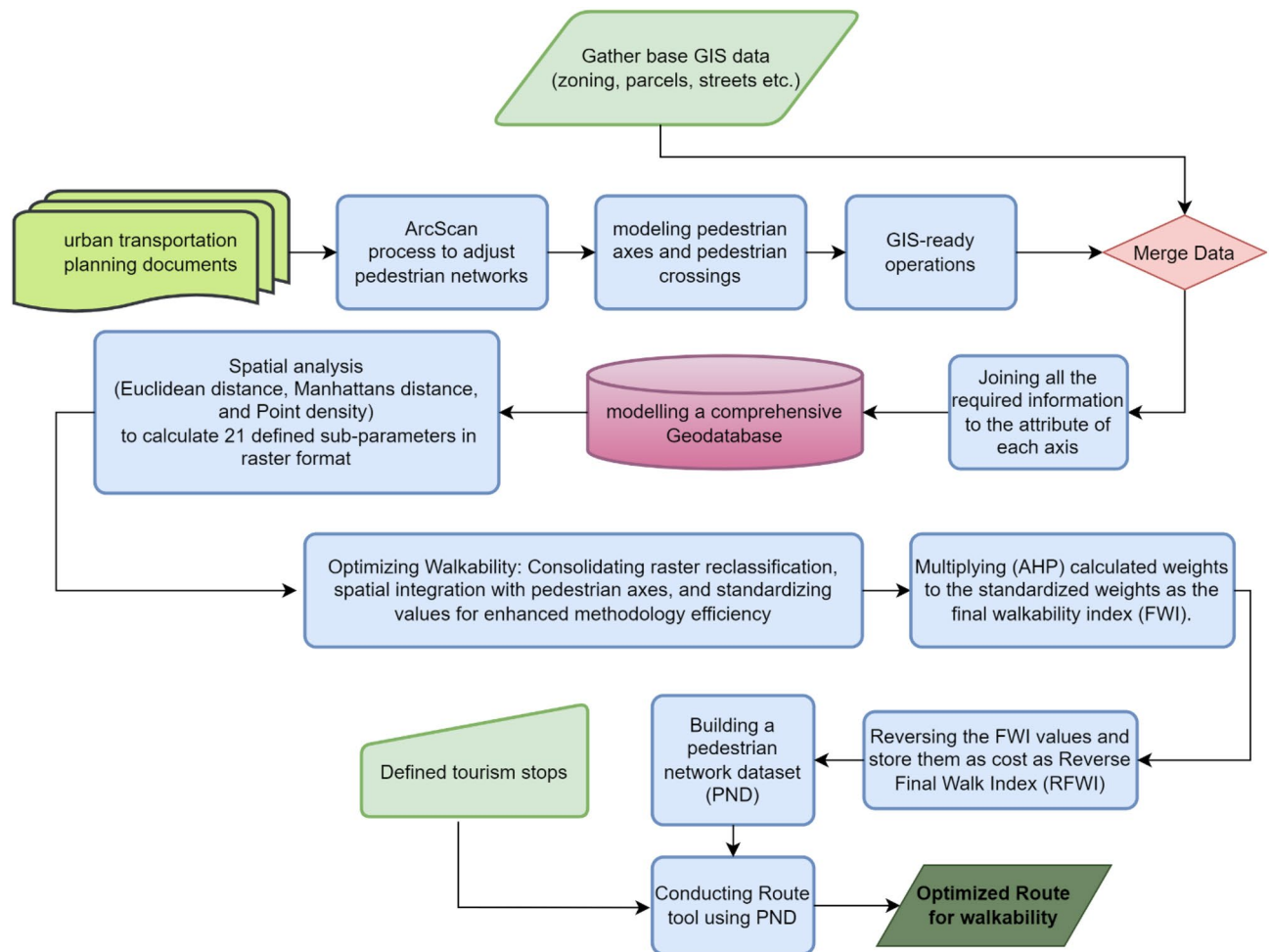


Figure 4. Optimizing urban walkability index as a tourism route flowchart.

routes by minimizing impedance and maximizing walkability index values assigned to network segments. The integrated dataset provided a robust common reference layer for overlaying multi-criteria walkability metrics within the GIS.

The shortest walking route between the identified attractions was also determined using Dijkstra's algorithm as a baseline for comparison (Fig. 8). This route optimization relies solely on minimizing travel distance, without incorporating impedance factors related to walkability.

Discussion

Our methodology differs from the shortest route algorithm approach utilized in certain articles as we employ GIS techniques and spatial analysis to comprehensively evaluate and optimize pedestrian infrastructure¹²³. By considering 21 indicators across five criteria categories, our approach surpasses the shortest route algorithm, providing a safer and more enjoyable walking experience for tourists. These indicators encompass factors such as infrastructure, historical preservation, cleanliness, aesthetics, and vegetation. While we acknowledge the significance of macro and micro indicators in assessing walkability²⁷, our study specifically focuses on evaluating and optimizing pedestrian infrastructure within historical cities. Using a multi-criteria GIS approach, our objective is to systematically and thoroughly evaluate walkability in urban environments, accounting for diverse factors that enhance both walkability and tourist satisfaction. We have prioritized references directly relevant to our research objectives and findings. Overall, our study contributes valuable insights to walking assessment tools, urban management decision-making, and the alignment of urban tourism, historical preservation, and pedestrian infrastructure development¹²³.

Recommendations include adopting a multi-criteria GIS approach to evaluate pedestrian networks in historic urban environments. This approach results in more effective tourist routes, considering various factors contributing to walkability and increasing tourist satisfaction. Urban management should prioritize the development of pedestrian infrastructure and urban design in important areas such as Jumhouri Street, Tabarsi Street, Enqelab Street, and Qaran Street. Investing in tourism facilities and supporting infrastructure in these areas generates positive impacts, including tourism growth, visitor satisfaction, increased revenues, job creation, and business revitalization.

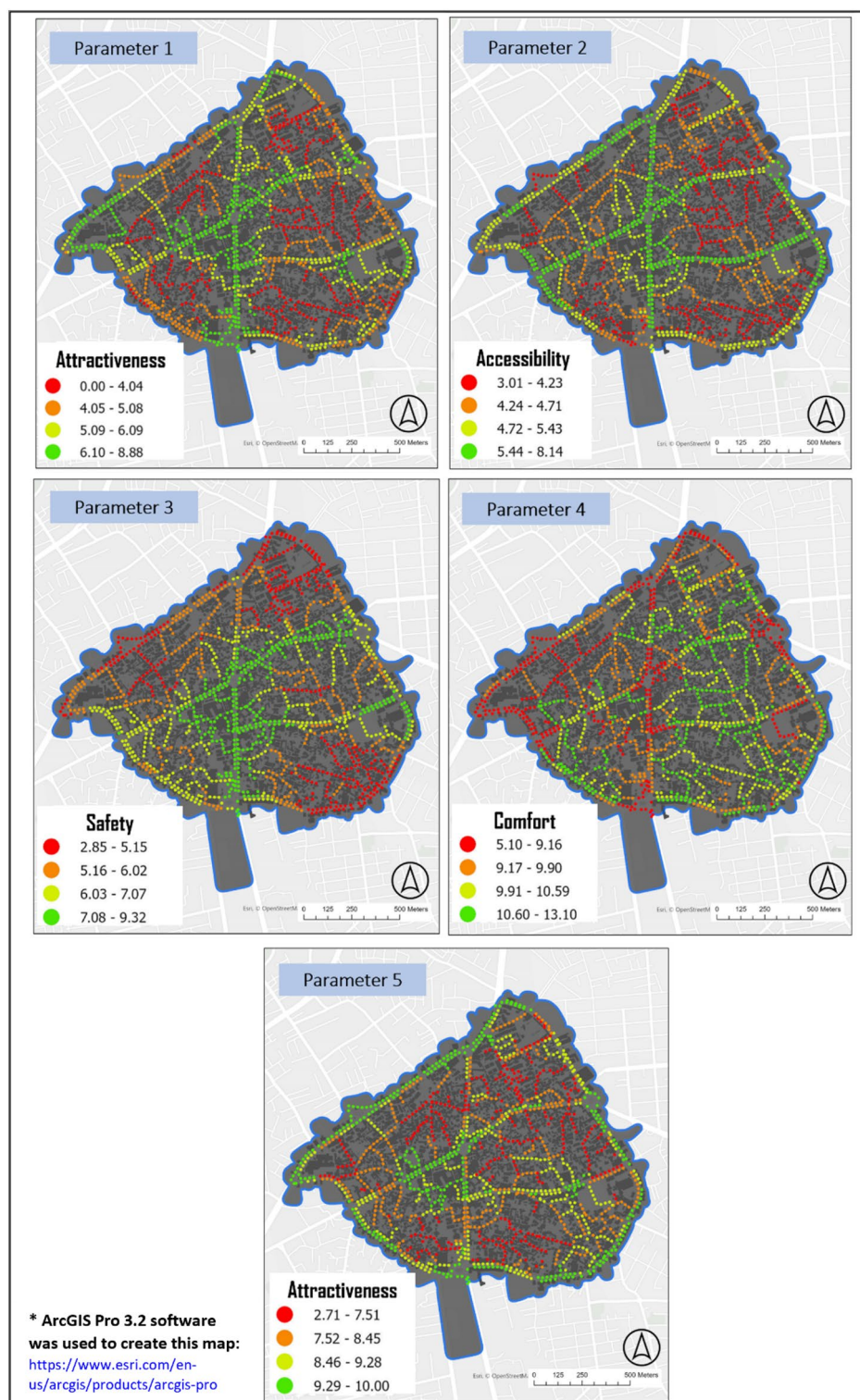


Figure 5. Overlapping sub-criteria and value recategorization maps for each criterion.

Aligning urban management, tourism development, and historical preservation objectives allows cities to capitalize on their cultural heritage and urban landscapes. By incorporating these considerations into development plans, cities can enhance pedestrian experiences, improve aesthetics, and boost tourism potential. Municipalities should allocate resources and implement strategies to enhance pedestrian infrastructure, leveraging the economic benefits associated with urban tourism. This proactive approach contributes to long-term economic growth and sustainability.

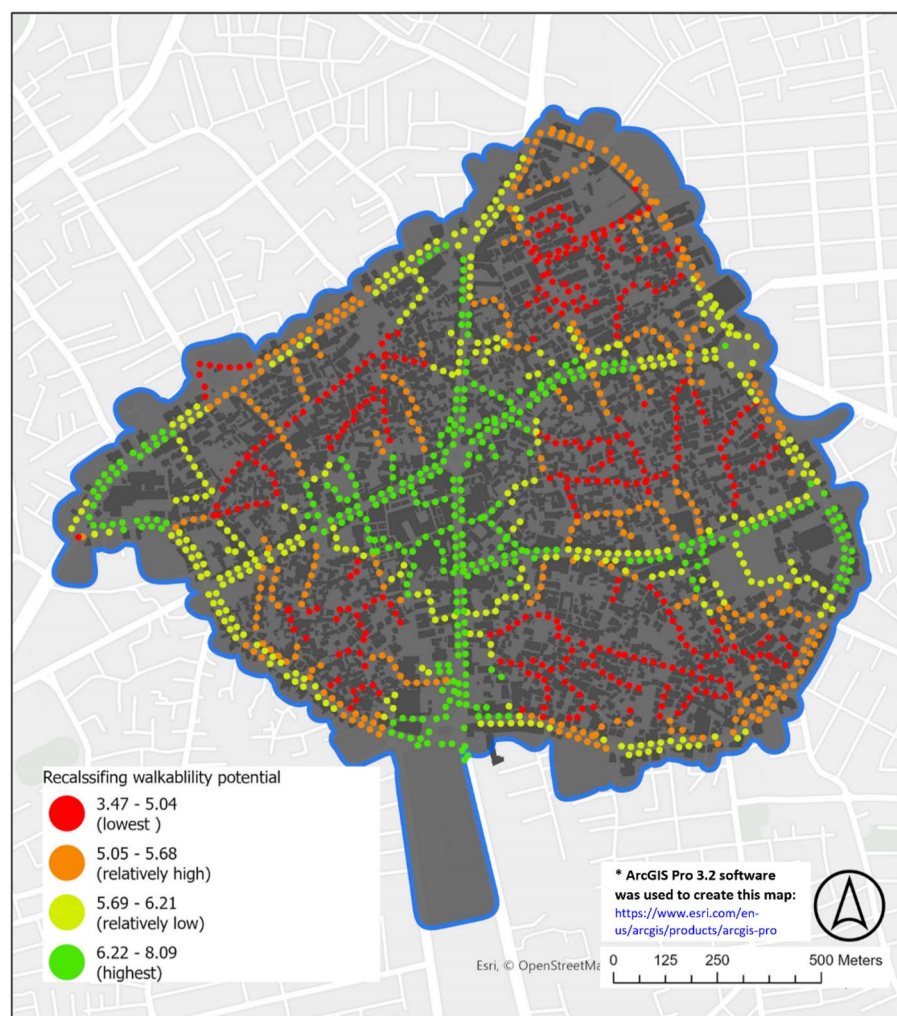


Figure 6. Final weighted overlap map of 5 key criteria in walkability assessment.

The efficacy of urban tourism in historical areas can be evaluated using various criteria and sub-parameters for assessing the walkability of the urban environment. The second tourist route in the historical area of Sari city, developed using the MCA approach, outperforms the first route, which was based solely on the shortest route algorithm.

The MCA approach considers 21 indicators across five criteria categories: attractiveness, accessibility, safety, comfort, and legibility, to evaluate the walkability of the urban environment. These indicators include the availability and distribution of non-residential land use, urban green spaces, public spaces, retail shops, urban furniture installations, sidewalk quality and width, physical barrier density, and proximity to public parking lots and transportation facilities, among others.

Thus, the comprehensive approach of the second route ensures that multiple factors contributing to the walkability of the urban environment are considered, providing a more enjoyable and safer walking experience for tourists. These positive experiences can lead to higher levels of satisfaction and favorable reviews, benefiting the tourism industry in the historical area of Sari City.

Additionally, to visually illustrate the superiority of our proposed methodology, we present a comparison between the optimal walking route, determined using the MCA approach, and the shortest walking route among the tourist attractions. This comparison is depicted in Fig. 9, which showcases the contrasting paths and highlights the enhanced experience offered by our approach in navigating the historical area of Sari city. Furthermore, Fig. 10 provides a statistical comparison of acquisition scores achieved in each modeling approach, offering quantitative insights into the effectiveness of our methodology compared to alternative approaches.

Figure 10 presents a statistical comparison of the acquisition scores achieved by the optimized route and the shortest route among tourist attractions. The optimized route, determined using the proposed multi-criteria approach, outperforms the shortest route in terms of various walkability indicators. The graph displays the mean, median, and standard deviation of the scores for each sub-parameter, highlighting the superior performance of the optimized route in providing a more pedestrian-friendly and enjoyable experience for tourists.

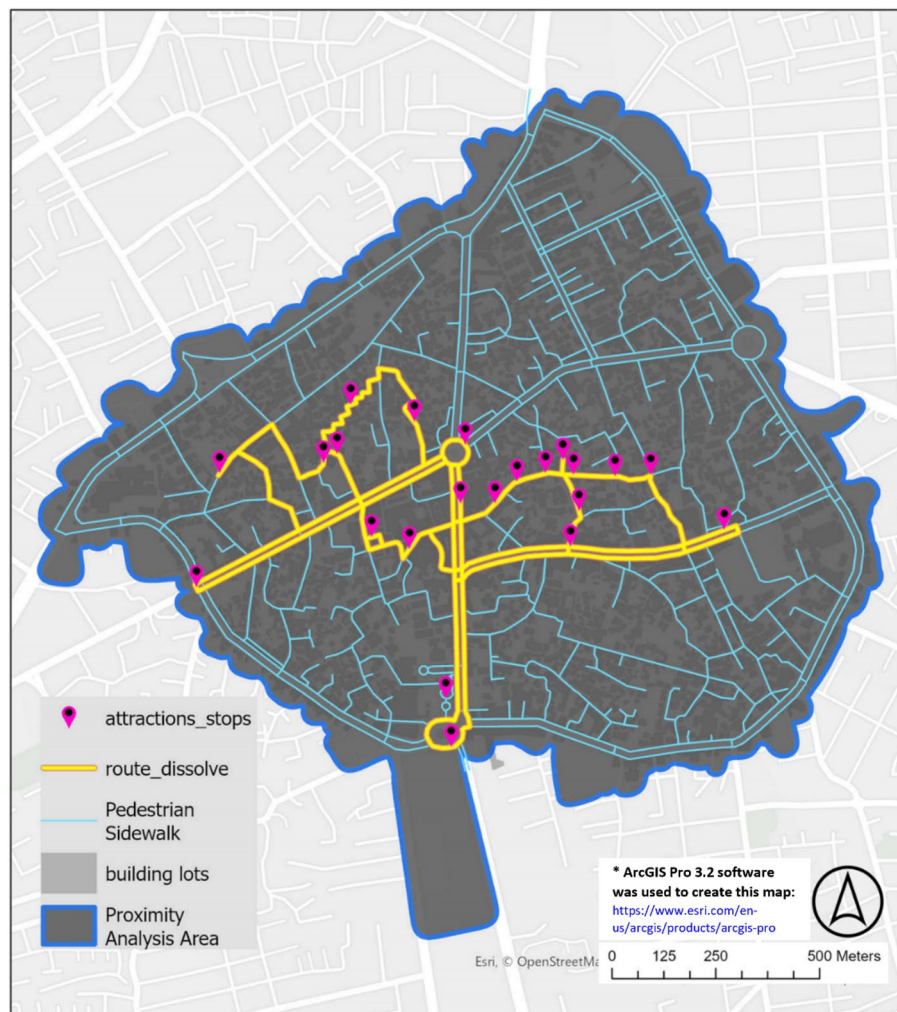


Figure 7. Optimal walking route based on the proposed methodology.

By conducting a comparative analysis between the "optimized" and "shortest route" options, it was observed that the utilization of the index maximization function (FWI) effectively enhanced the walkability along the proposed route. In Fig. 8, the cumulative score of the edges along the selected Route reached 1676 points, which exceeded the score obtained from the shortest Route function by 172 points, representing a difference of approximately 11.5%. The graphical representation of mean, median, and standard deviation dispersion indices further emphasize the noticeable superiority of the edges in the first Route compared to the second Route. This significant improvement in walkability demonstrates the successful application of the FWI optimization approach in enhancing the overall pedestrian experience and accessibility along the proposed route. Despite the longer distance (Selected Route total length = 6.1 km > Shortest Route total length = 5.2 km), prioritizing the walking index in urban tourism and historical city contexts can lead to significant economic benefits. In Table 4, a comprehensive statistical analysis compares the optimized and shortest route options based on walkability indicators and criteria within the study. The impedance-optimized route outshines its counterpart across various factors, showcasing superior scores in attractiveness, accessibility, safety, comfort, and legibility. The optimized path excels in a wide range of built environment metrics, including land use mix, public space, green space, sidewalk width, lighting, traffic management, and intersection density. This substantiates that prioritizing walkability criteria over mere distance minimization identifies a significantly more pedestrian-friendly pathway connecting tourist attractions.

Despite its modestly increased total length, the index-maximized route offers a notably enhanced walking experience, as outlined in Table 4's statistical comparison. The optimized pathway excels in all five walkability criteria and a majority of sub-parameters, emphasizing improvements in amenities, safety, aesthetics, and pedestrian infrastructure. This results in a more gratifying tourist experience, even with slightly extended travel time. Recognizing that journey fulfillment often surpasses brevity in tourism goals, the optimized route, enriched by walkability indicators, crafts a more satisfying and engaging path connecting heritage sites. In Sari's historic

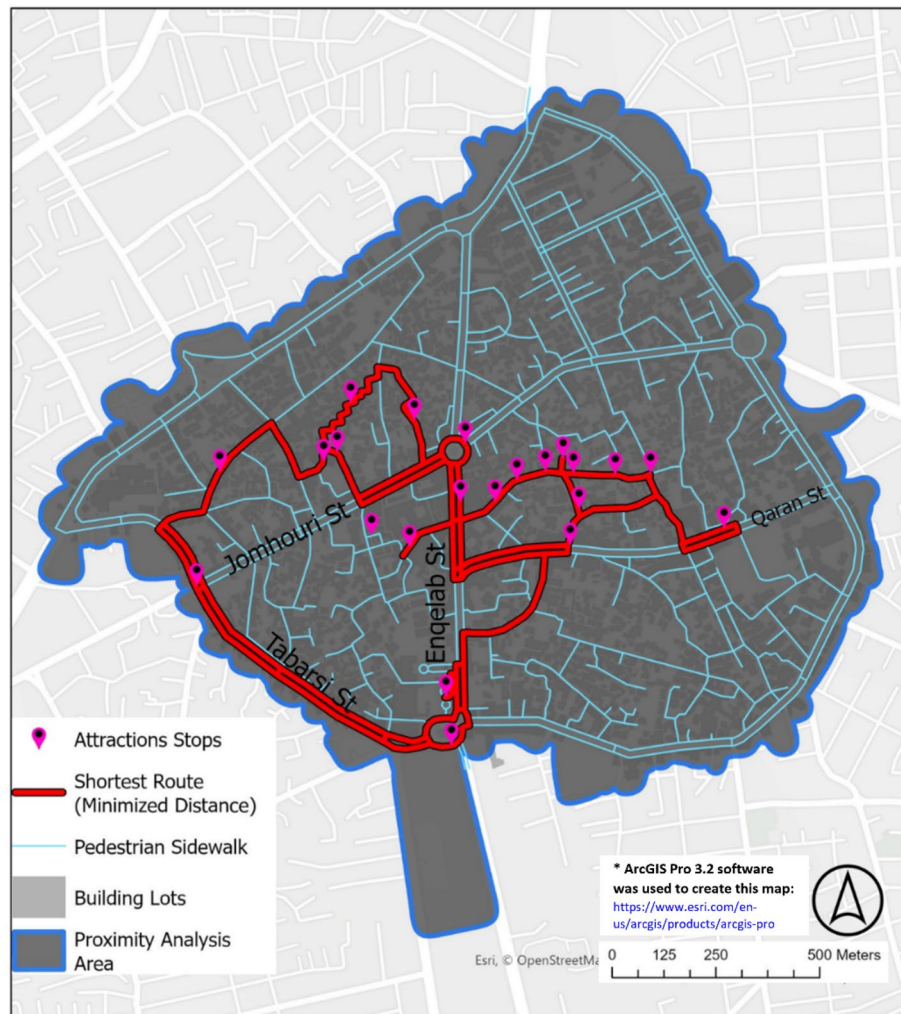


Figure 8. Shortest walking route between tourist attractions based on Dijkstra's algorithm.

core, this pedestrian-friendly route, shaped by multi-criteria optimization, emerges as an ideal strategy for urban tourism, promising visitors a more profound experience of the cultural landscape.

The results presented in this study highlight the importance of considering various factors in assessing walkability and optimizing tourist routes in historical cities. The proposed methodology allows for a comprehensive evaluation of different indices, each representing a specific aspect of the built environment that contributes to walkability.

Among the sub-parameters examined, some stand out as particularly significant for historical cities like Sari. For instance, the land use mix, availability of public spaces, and access to green areas play a crucial role in creating an attractive and enjoyable environment for pedestrians and tourists alike. Similarly, the condition and width of sidewalks, as well as the presence of obstacles, directly impact accessibility and comfort levels.

While the proposed methodology provides a robust framework for assessing walkability and optimizing tourist routes, it is important to acknowledge some potential limitations and areas for future research. One limitation lies in the subjectivity involved in the weight calculation process using AHP. Although efforts were made to ensure consistency, the inclusion of additional stakeholders or experts in the pairwise comparisons could lead to different weight assignments.

Furthermore, the current study focuses on the physical attributes of the built environment, but it could be enriched by incorporating additional factors such as cultural elements, social dynamics, and economic considerations. Future research could explore the integration of these aspects to provide a more holistic understanding of walkability and tourism in historical contexts.

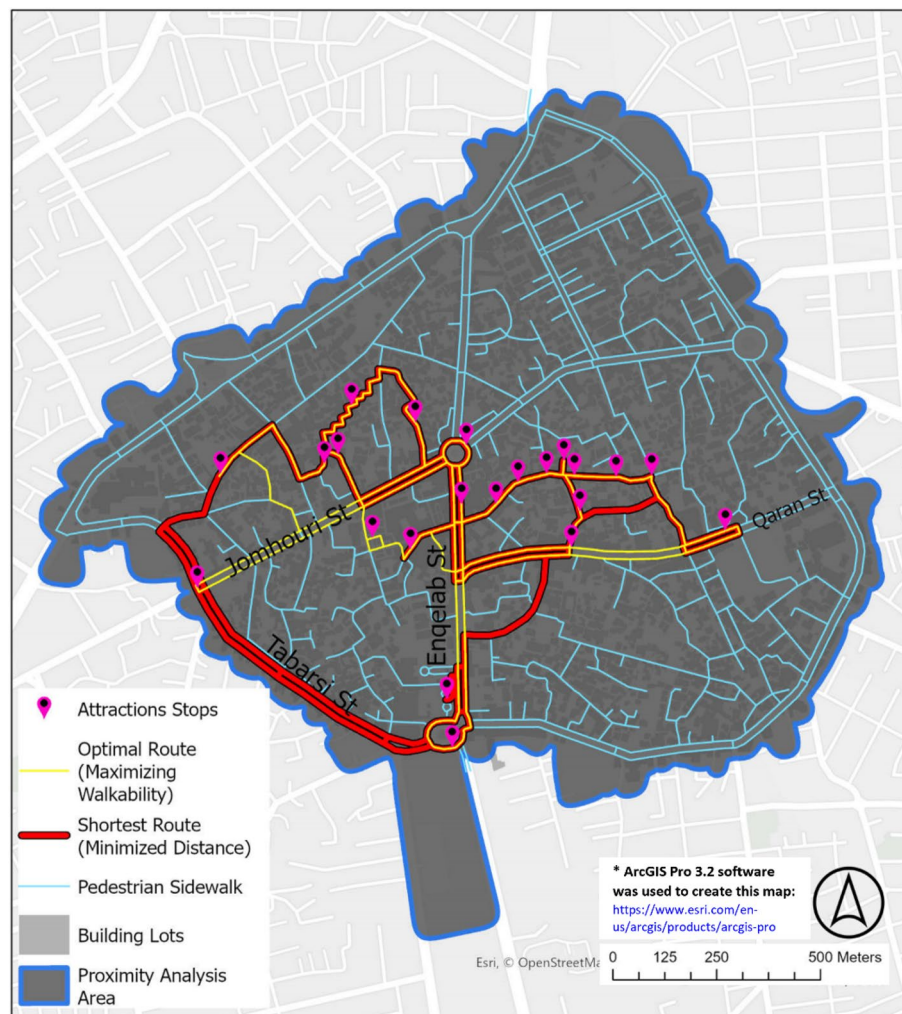


Figure 9. Comparison of optimal walking route and shortest walking route among tourist attractions.

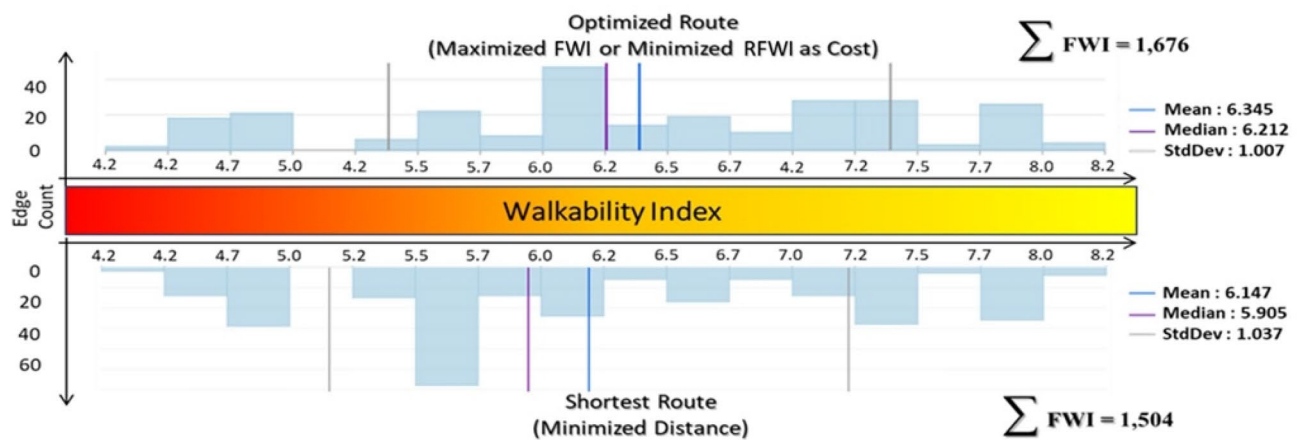


Figure 10. Statistical comparison of acquisition scores in each modeling approach.

Indicators	Optimized route scores	Shortest route scores
Sub-parameters		
Land use-Mix	7.796	7.269
Retail	5.445	5.590
Public space	3.759	2.838
Green space	5.499	3.993
Furniture	1.660	1.436
Sidewalk Pavement condition	6.424	6.045
Sidewalk width	6.888	5.294
Obstacle	8.398	7.798
Proximity to public parking lots	7.052	6.177
Proximity to public transportation	2.507	2.469
Proximity to police stations	7.954	7.011
Lighting system	5.456	5.090
Proximity to crosswalk	5.456	4.542
Traffic management	8.367	5.728
Intersection density	5.834	5.088
Shade	6.195	5.751
Shelter	2.761	2.294
Noise	7.050	7.207
Smell	9.502	7.384
Permeability	8.806	6.534
Enclosure	8.524	8.462
Parameters		
Attractiveness	6.039	5.444
Accessibility	5.634	5.116
Safety	6.556	5.625
Comfort	7.430	6.653
Legibility	8.735	7.016
Total length (km)	6.1	5.2

Table 4. Statistical comparison of the key indicators between the two routes.

Conclusion

This research demonstrates the efficacy of an impedance-optimization approach for improving walkability and tourism accessibility in historical cities, answering three key questions: First, the GIS pedestrian routing model developed enables the identification of optimal paths connecting heritage sites while avoiding impedance factors, enhancing accessibility (RQ1). Second, the multi-criteria evaluation framework identifies key built environment indicators that should be prioritized in modeling walkable tourist routes (RQ2). Finally, the impedance optimization methodology provides multidimensional benefits including improved infrastructure, boosted tourism, and preserved cultural heritage (RQ3).

The methodology introduces an innovative framework using integrated AHP-GIS analysis to assess various built environment and streetscape indicators crucial for walkability. Illustrated by the case study of Sari, Iran, this technique facilitates identifying optimal pedestrian routes connecting heritage sites while avoiding factors hindering walkability. Findings suggest historical cities can notably enhance tourism accessibility by employing GIS optimization to determine efficient, pedestrian-friendly paths between attractions. A significant contribution is identifying 21 key walkability indicators across five criteria: attractiveness, accessibility, safety, comfort, and legibility, supported by a literature review synthesizing pertinent built environment factors. The subsequent AHP-GIS assessment underscores the value of evaluating diverse walkability criteria to maximize pedestrian friendliness. Urban planners and tourism developers are urged to adopt this methodology, integrating multiple walkability metrics in tourism route design. The impedance-based optimization approach showcases potential to enrich the tourism experience, stimulate economic growth, and safeguard cultural heritage. While geographically limited to Sari, Iran, the proposed framework offers a transferable model for enhancing walkability in historical cities globally. This methodology serves as a valuable tool for urban planning and tourism design, with long-term validation across diverse contexts imperative for sustainability.

Data availability

The data is available from the corresponding author.

Appendix

See Fig. 11.

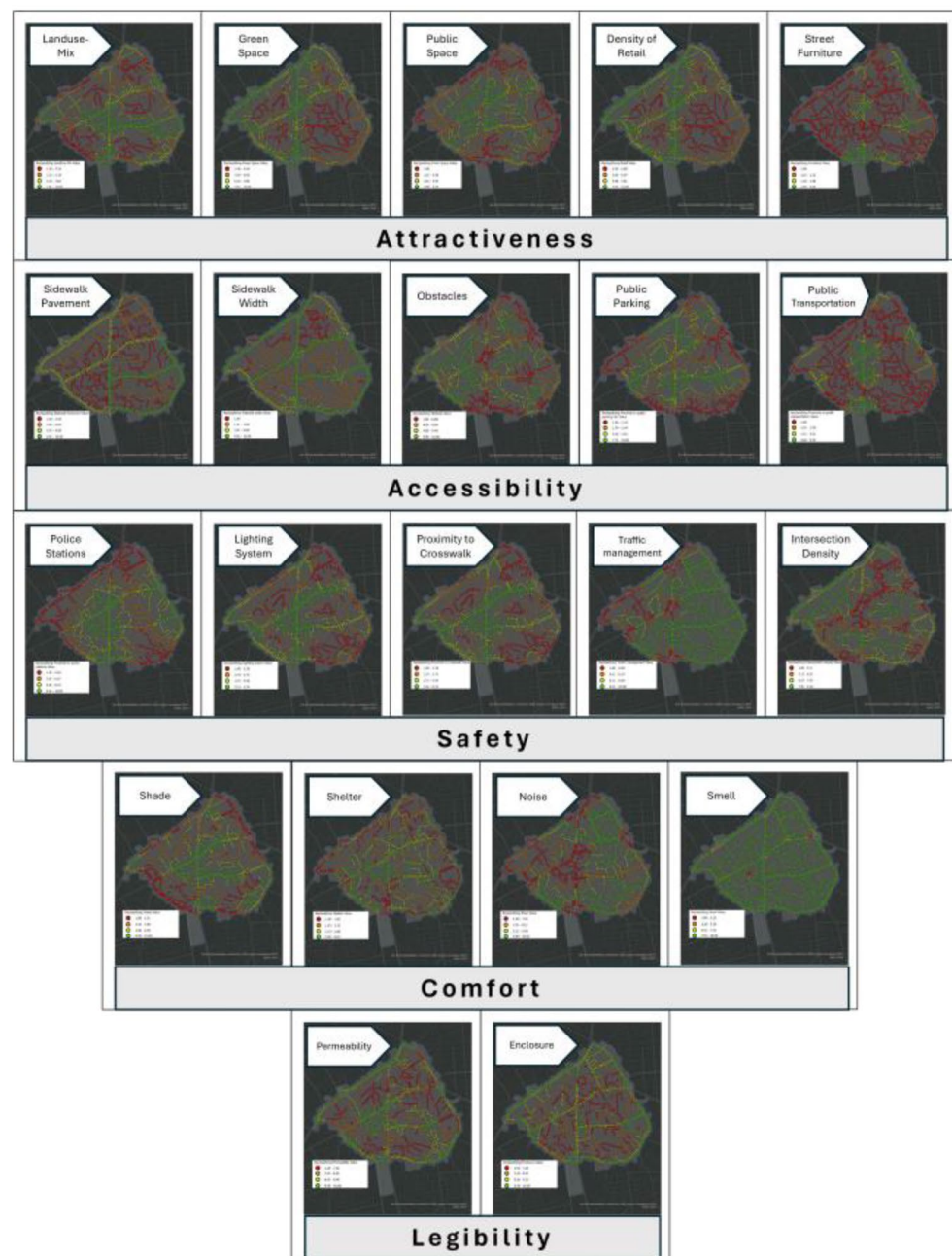


Figure 11. Walkability index evaluation based on 21 sub-parameters.

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Author contributions

OM: Conceptualization; Data collection; investigation; data analysis; visualization; writing; formatting. AS: Ali Soltani: Supervision; writing; editing. MRQ: Project administration; data analysis; writing.

Competing interests

The authors declare no competing interests.

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